MORGAN STATE UNIVERSITY

Office of Procurement · HOEN Site, Suite 1202 · 1700 E. Cold Spring Lane · Baltimore, Maryland 21251

Performance Work Statement

Professional Structural Engineering Services

Design Verification, Certification and Sealing of the Reinforced-Concrete Foundation, Anchorage, Drainage and Spill Containment for a Rocket-Engine Static-Fire Test Stand

Lake Clifton Aerospace Engine Test Site · Baltimore, Maryland

Solicitation number	_____ ☐ RFP (Best Value) ☐ RFQ ☐ Small procurement ☐ Cooperative
Requisition number	_____
PWS document number	PWS-LC-FND-001
Revision	3 — Issued for University review and solicitation
Supersedes	Rev. 2 (September 2026, not released); Rev. 1 (August 2026); Rev. 0A and Rev. 0 (technical design basis)
Prepared	5 November 2026
Date of issue	_____
Owner / requesting department	Morgan State University (§1.4) — School of Engineering, Aerospace Engineering Program
Contract type	Firm-fixed-price professional engineering services, priced by deliverable (Appendix C). No construction is procured under this PWS.
Governing procurement policy	Morgan State University Procurement Policies and Procedures, Appendix A Terms and Conditions incorporated by reference (§11.2); Morgan State University Construction General Conditions (2022) as referenced in §3.6 and §3.7
Governing codes	ACI 318-19; ASCE/SEI 7-22; IBC 2021 Chapter 17; NFPA 30; COMAR 26.10 (as applicable)

Solicitation review and approval

Role	Name / Office	Signature	Date
Requestor	Mazen Badr — Aerospace Engineering		
Technical reviewer	Consolata Irungu — Aerospace Engineering		
Principal Investigator / Faculty advisor	Dr. Oluwatobi Busari — School of Engineering		
Project Manager	Angelo Pirali		

Role	Name / Office	Signature	Date
Procurement Officer	Oluwagbenga Alo — Office of Procurement		
MBE Liaison Officer	Karen Reyes		
Design and Construction Management			
Facilities Management			
Office of Safety, Health and Environment			
Administrative / legal review			

Revision History

Rev.	Date	Description of change	Prepared by
0	July 2026	Internal engineering design report for the foundation and anchorage, issued for Professional Engineer review.	Aerospace Engineering
1	August 2026	Rewritten as a contractor-facing PWS. Engineering carried as Appendix A; open items in Appendix B.	Aerospace Engineering
0A	September 2026	Technical correction of the design basis: anchor specified as the Hilti HDA-TR M12 stainless undercut anchor per ICC-ES ESR-1546; anchor, stability and pad calculations recalculated; dynamic and flame-deflector checks added; exposure class F2.	Aerospace Engineering
2	September 2026	Office of Procurement comments of 3 and 4 September 2026: ownership and site, definitions, allocation of responsibilities, and insurance, indemnification, bonds and solicitation notices quoted verbatim from the Policies. Not released.	Aerospace Engineering
3	November 2026	Combines Rev. 2 and Rev. 0A and adds the drainage and spill-containment design (level pad, sloped apron, trench drain, buried containment tank), the risk-allocation and non-interference requirements of the Office of Procurement email of 7 October 2026, itemized pricing by deliverable, and the firing parameters of PWS-LC-FPE-001 Rev. 1. See Summary of Revision 3 Changes.	Aerospace Engineering

Summary of Revision 3 Changes

Each row states the source of the change and where it is answered in this revision.

No.	Source	Revision 3 response	Where
1	Office of Procurement, 7 October 2026: primary responsibility and risk on the vendor; vendor solely responsible for methods, safety and compliance with laws, including state environmental regulations and campus policies, and indemnifies the University (Morgan State University Construction General Conditions (2022) §1.02 and §6.01).	New §3.6 places the risk of each scope on the party performing it: professional design risk on the Contractor, construction risk on the Construction Contractor. Environmental compliance is a design requirement (Task 6). The University assumes no indemnity (§11.4).	§3.3, §3.6, §7.6, §11.4
2	Office of Procurement, 7 October 2026: non-interference with existing site operations and other active projects (§4.03 Adjacent Work, §6.07 Separate Contracts).	New §3.7 requires coordination with the fire-protection engineering work (PWS-LC-FPE-001), the test-stand fabrication, and any other University project at the site, with no disruption to or damage of existing facilities.	§1.6, §3.7, §12
3	Office of Procurement, 3 November 2026: itemized quotation supporting price reasonableness (COMAR 21.05.07; General Conditions §3.06 and §3.07).	Appendix C price schedule: one price per deliverable and per option. Payment on written acceptance of each priced deliverable replaces the Rev. 2 percentage milestones.	§14, App. C
4	Office of Procurement, 3 August 2026: percentage milestones may not be necessary for professional services.	Milestone percentages removed (see No. 3).	§14

No.	Source	Revision 3 response	Where
5	Principal Investigator, 21 October 2026 (conditions for the fire-protection piggyback): MSU contract terms, additional insured, XCU coverage, itemized price by deliverable.	Applied to this PWS: MSU terms govern (§11.2); MSU additional insured (§11.3 item 3); XCU coverage required for any subsurface work and flowed to the construction contract (§11.3.1); itemized prices (App. C).	§11, App. C
6	Technical design basis Rev. 0A (29 September 2026).	Appendix A replaced with the Rev. 0A design: 16 Hilti HDA-TR M12x125/30 stainless anchors, h_ef = 125 mm, baseplates bearing directly on concrete, exposure F2, recalculated checks. The ⅝ in, 10 in embedment anchor of Rev. 1 and Rev. 2 is withdrawn.	App. A
7	Drainage concept sent to the Principal Investigator on 19 August 2026 (level pad, sloped apron, trench, closed tank); buried containment tank adopted by the design team in November 2026.	Level 8 × 8 ft deck, 18 in apron at ½ in per ft, trench drain with steel grate, outlet dropping into a buried double-wall containment tank. New Task 6 and deliverable D-7.	§3.1, §7.6, App. A.6
8	PWS-LC-FPE-001 Rev. 1 (fire protection) firing parameters.	Firing durations aligned: ignition-verification firings of about 5 s, sustained firings to depletion of about 12.4 s, 30 s design envelope. On-stand propellant 38.5 lbm RP-1 and 99.4 lbm LOX.	§1.1, §7.5, App. A
9	Test-stand fabrication by McKinney Corp; question of an out-of-state PE (30 October 2026).	The Engineer of Record for the foundation must hold a Maryland license at the time of sealing; licensure by comity is acceptable. The stand is owner-furnished equipment; its fabricator is a separate contract (§1.5, §3.7).	§1.5, §6

Table of Contents

1 Introduction and Background
2 Purpose
3 Scope of Services and Allocation of Responsibilities
4 Period of Performance
5 Place of Performance
6 Contractor Qualifications and Key Personnel
7 Tasks
8 Deliverables and Delivery Schedule
9 Performance Requirements Summary
10 University-Furnished Information and Services
11 Contractor-Furnished Items, Insurance and Risk Allocation
12 Site Access, Safety and Security
13 Meetings, Reporting and Communication
14 Acceptance and Payment
15 Applicable Codes and Standards
16 Points of Contact
17 Attachments
Appendix A — Technical Design Basis and Calculations
Appendix B — Open Items to be Closed by the Contractor
Appendix C — Itemized Price Schedule
Appendix D — Preliminary Drawings

1 Introduction and Background

1.1 The facility

Morgan State University, School of Engineering, Aerospace Engineering Program, is establishing a liquid rocket-engine static-fire test capability at the former Lake Clifton High School property in Baltimore, Maryland. The facility consists of a steel test stand mounted on a reinforced-concrete foundation. The engine is mounted vertically and fires downward, so the thrust reaction acts upward and places the foundation anchorage in tension. Maximum design thrust is 3,000 lbf.

The engine burns RP-1 (kerosene) with liquid oxygen (LOX). For each firing the stand holds about 38.5 lbm of RP-1 and 99.4 lbm of LOX. The program consists of ignition-verification firings of about 5 seconds and sustained firings to propellant depletion of about 12.4 seconds; a 30-second firing envelope is used for thermal and containment design, consistent with the fire-protection Performance Work Statement PWS-LC-FPE-001 Revision 1.

1.2 The preliminary design

The University's design team, under faculty supervision, has developed a preliminary design of the foundation, anchorage, drainage and spill containment, with structural calculations to ACI 318-19. The design is provided in Appendix A and the drawings in Appendix D as the technical basis for the services solicited here. Appendix A and Appendix D are furnished for information only; they are not a certified design and the University does not warrant them.

1.3 The services required

The University requires the services of a licensed Professional Engineer to independently verify that design, complete the analyses that remain open, and certify and seal the foundation, anchorage, drainage and spill-containment design so that construction and test operations may proceed. No test firing will be conducted until a sealed certification has been received and accepted.

1.4 Ownership and site

The Lake Clifton Aerospace Engine Test Site is real property of Morgan State University. The reinforced-concrete foundation, the drainage and containment system, the steel test stand, the engine and all associated equipment are, and upon completion will remain, assets of the University. No faculty member, student, vendor or other party holds or will acquire any ownership interest in the facility or any part of it.

The University, acting through the Office of Procurement, is the Owner for the purposes of this PWS and of the separately procured construction contract. The Principal Investigator identified in §16 is the University's technical representative and acts on the University's behalf; that role does not confer ownership of, or personal responsibility for, the facility. Control, operation, maintenance and inspection of the facility after acceptance rest with the University through the Principal Investigator, Facilities Management and the Office of Safety, Health and Environment, in accordance with University policy.

1.5 Definitions

Term	Meaning
University; Owner	Morgan State University, the owner of the site and the facility and the contracting party under this PWS and under each separate contract.

Term	Meaning
Contractor	The professional engineering firm awarded a contract under this PWS. Its services are professional engineering services only. "Engineer" or "Engineer of Record" means the Contractor's Professional Engineer, licensed in Maryland, responsible for the sealed design.
Construction Contractor	The firm separately procured by the University to construct the foundation, drainage and containment system and to set the test stand. It is not a party to any contract under this PWS, and the Contractor has no contractual relationship with it.
Stand Fabricator	The firm fabricating and assembling the steel test stand under its own contract with the University (currently McKinney Corp). It is not a party to any contract under this PWS.
Principal Investigator (PI)	The University faculty member responsible for the technical direction of the test program, identified in §16. The PI is a University employee acting for the University and is not an owner of, or a party to, any contract under this PWS.
General Conditions	The Morgan State University Construction General Conditions (2022), which govern the construction contract and are referenced in §3.6 and §3.7.
Adjacent work	Any other work, project or operation of the University or its separate contractors at or near the site, including the work listed in §1.6.
Special Inspector; testing agency	The accredited agency performing special inspections and materials testing under IBC 2021 Chapter 17, engaged through the Contractor under Task 8 or directly by the University.
Notice to Proceed (NTP)	The written notice issued by the Office of Procurement authorising the Contractor to begin work, from which all durations in this PWS are measured.

1.6 Related work at the site

The following work is being carried out by the University under separate contracts. The Contractor shall coordinate with it under §3.7 and shall not be responsible for it.

- Fire protection engineering services for the test site under PWS-LC-FPE-001 Revision 1, including the quantity-distance analysis, fire-protection and water-spray design, and Baltimore City Fire Marshal approval.
- Fabrication, shop assembly and delivery of the steel test stand by the Stand Fabricator.
- Test-program operations by the Principal Investigator's team, and any other University project active at the Lake Clifton property during the period of performance.

2 Purpose

This PWS defines the professional structural engineering services required to produce a signed and sealed design certification for the reinforced-concrete foundation, anchorage, drainage and spill-containment system described in Appendix A, with the supporting analyses, drawings, specifications and optional construction-phase engineering services.

This PWS is written in performance terms. It states the outcomes the Contractor shall achieve and the standards they shall meet. Except where a specific code provision, material or product is identified, the Contractor selects the analytical methods and professional techniques used to achieve the required results.

This PWS procures professional engineering services only. It does not procure construction, fabrication, installation, materials or equipment. Construction will be procured by the University under a separate solicitation governed by the General Conditions. This solicitation is governed by

the Morgan State University Procurement Policies and Procedures, whose Appendix A terms and conditions are incorporated by reference (§11.2); **submission of a response constitutes acceptance of those terms.**

3 Scope of Services and Allocation of Responsibilities

3.1 Services included

- Independent review and verification of the University's foundation and anchorage design and calculations in Appendix A.
- Wind-load analysis to ASCE/SEI 7-22 and the resulting load combinations.
- Geotechnical review: allowable bearing pressure, settlement, friction coefficient and required subgrade preparation over the existing fill, including the effect of the containment-tank excavation on the foundation.
- Verification of the anchorage to ACI 318-19 Chapter 17 for all applicable failure modes, including confirmation of the Hilti HDA-TR M12 against ICC-ES ESR-1546 and its acceptance for rocket-firing loads.
- Structural check of the test-stand baseplate connection, and review of the flame deflector, its anchorage and the thermal protection of the concrete.
- Design of the drainage and spill-containment system: level deck, sloped apron, trench drain, outlet, buried containment tank and its hold-down, including identification of the environmental regulations and approvals that apply to it (Task 6).
- Signed and sealed drawings, calculations and specifications suitable for construction, and a statement of special inspections to IBC 2021 Chapter 17.
- At the University's option, construction-phase engineering services limited to submittal review, responses to requests for information, special inspection and a final conformance statement (Task 8).

3.2 Services excluded

- **Construction** of the foundation, drainage and containment system, including all construction means, methods, techniques, sequences and procedures, and all construction site-safety programs and precautions. These belong to the Construction Contractor under the General Conditions.
- **Fabrication, assembly, delivery, erection, maintenance or modification of the steel test stand**, which is University property furnished to the Contractor as design data only (§10).
- Design of the steel test-stand structure above the baseplates.
- Design of propellant storage, feed, ignition, control, range-safety and fire-protection systems (fire protection is procured under PWS-LC-FPE-001).
- Hazardous-materials assessment or abatement of existing site conditions.
- Subsurface drilling and laboratory testing, unless the Contractor proposes it as a priced option under Task 3.
- Supervision, direction or control of the Construction Contractor, the Stand Fabricator or their subcontractors, or responsibility for their acts or omissions.
- Ownership of, or operational responsibility for, any part of the facility.

3.3 Allocation of responsibilities

R — responsible for the function and its adequacy; **A** — reviews, accepts or approves for the University; **S** — furnishes information, access or support; **—** — no responsibility. Functions of the Construction Contractor and the Stand Fabricator are governed by their separate contracts and are shown only to bound the Contractor's scope.

Function	University (Owner)	Contractor (Engineer)	Construction Contractor	Stand Fabricator	PI
Ownership of the site, foundation, containment system, stand and equipment	R	—	—	—	—
Preliminary design basis (Appendix A), for information only	S	A [1]	—	—	S
Professional design and structural adequacy of foundation and anchorage	A	R	—	—	S
Design of drainage and spill containment, including tank sizing and hold-down	A	R	—	—	S
Identification of environmental regulations and approvals for the designed work	A	R	—	—	S
Application for permits and registrations where the law requires the Owner to apply	R [2]	S	S	—	S
Signed and sealed drawings, calculations, specifications and certification	A	R	—	—	S
Statement of special inspections	A	R	—	—	—
Solicitation, award and administration of this contract and of the construction contract	R	—	—	—	S
Fabrication, assembly and delivery of the test stand; baseplate hole size	A	— [3]	—	R	S
Construction: means, methods, techniques, sequences and procedures	—	—	R	—	—
Compliance with laws, codes, environmental regulations and campus policies during construction	—	—	R	—	—
Construction site safety and OSHA compliance	—	— [4]	R	—	—
Setting and anchoring the stand on the completed foundation	—	—	R [5]	S	S
Coordination with adjacent work and protection of existing facilities	A	R [6]	R	R	S
Construction submittals and requests for information	A	R [7]	R	—	S

Function	University (Owner)	Contractor (Engineer)	Construction Contractor	Stand Fabricator	PI
Special inspections and materials testing during construction	A	R [7]	S	—	—
Final statement of conformance of the constructed work	A	R	S	—	—
Insurance and indemnification	Per policy	R [8]	R [9]	R [9]	—
Operation, maintenance, containment-tank pump-out and disposal after acceptance	R	—	—	—	R [10]
Test operations, range safety and engine firing	R	—	—	—	R [10]
Review and acceptance of deliverables; payment	R	—	—	—	S [11]

[1] Verifies independently; does not rely on it without verification. [2] Through the Office of Safety, Health and Environment. [3] Specifies the required baseplate holes and thickness (Task 5). [4] Own personnel only (§12). [5] Or the Stand Fabricator, as the University directs. [6] For its own services and site visits. [7] Under Task 8 if exercised. [8] Per §11.3 and §11.4. [9] Per its separate contract. [10] Acting for the University under University policy. [11] Technical recommendation to the Procurement Officer.

3.4 Limitation of the Contractor's responsibility

The Contractor shall not have control over or charge of, and shall not be responsible for, construction means, methods, techniques, sequences or procedures, or for safety precautions and programs in connection with construction, all of which are solely the responsibility of the Construction Contractor under its separate contract. The Contractor shall not be responsible for the acts or omissions of the Construction Contractor, the Stand Fabricator or their subcontractors.

Site visits, special inspections and construction-phase services under Task 8 determine whether the constructed work conforms to the sealed design and the applicable code. They do not constitute supervision of the Construction Contractor's work and do not relieve it of responsibility for that work.

Nothing in this section limits the Contractor's full responsibility for the professional adequacy of its own design, analyses, drawings, specifications, inspections and certification, its compliance with law in performing them, or its obligations under §3.6 and §11.

3.5 The University's role

The University's obligations under this PWS are those of an Owner procuring professional services: furnishing the information listed in §10, providing site access, reviewing and accepting deliverables, and paying for accepted work. The University does not warrant furnished information, and the Contractor shall independently verify any furnished information relied upon in its sealed design. Nothing in this PWS makes the University, the Principal Investigator or any University employee responsible for the professional adequacy of the Contractor's design or for the performance of the Construction Contractor or the Stand Fabricator.

3.6 Risk allocation and compliance

Consistent with the Morgan State University Construction General Conditions (2022), Sections 1.02 and 6.01, as cited by the Office of Procurement on 7 October 2026, the primary responsibility and risk associated with each part of the project rest with the vendor performing it, not with the University:

- **The Contractor** bears the risk of its professional services. It shall comply with all federal, State, local and campus laws, codes, regulations and policies applicable to those services, including Maryland environmental regulations on land, air and water pollution control as they affect the work it designs. It shall design the drainage and containment system so that no quench water, wash water or propellant from the pad is discharged to the ground, groundwater or the storm-drain system (Task 6).
- **The Construction Contractor** is solely responsible, under the General Conditions, for construction methods, safety and compliance with all applicable laws, codes, environmental regulations and campus policies during construction, and shall defend, indemnify and hold harmless the University against claims arising from its failure to meet those obligations. The construction solicitation will state these terms; this PWS requires the Contractor's specifications to carry them.
- **The University** assumes no obligation to indemnify any party (Policies, Appendix A, Section D, item 23, quoted in §11.4).

> *Office of Procurement: please insert or attach the exact text of General Conditions Sections 1.02 and 6.01 here if it is to be quoted in the solicitation.*

3.7 Non-interference with existing site operations and adjacent work

Consistent with General Conditions Section 4.03 (Adjacent Work) and Section 6.07 (Separate Contracts), as cited by the Office of Procurement on 7 October 2026, the Contractor shall cooperate fully with existing site operations and with every other active project at the site, including the work listed in §1.6. In particular the Contractor shall:

- schedule site visits through the University point of contact so that they do not conflict with test operations, the fire-protection engineer's work or other University projects;
- not disrupt, damage or obstruct existing facilities, utilities, paving, fencing or equipment at the site;
- coordinate its design with the fire-protection design under PWS-LC-FPE-001, in particular the water-spray and deluge flows that the drainage system must carry, the separation distances that govern the containment tank and trench location, and the deflector exhaust direction;
- coordinate the baseplate requirements of Task 5 with the Stand Fabricator through the University point of contact, so that the stand is delivered with baseplates that suit the selected anchor;
- require its specifications to place the same obligations on the Construction Contractor.

> *Office of Procurement: please insert or attach the exact text of General Conditions Sections 4.03 and 6.07 here if it is to be quoted in the solicitation.*

4 Period of Performance

The base period of performance is one hundred twenty (120) calendar days from the Notice to Proceed, for Tasks 1 through 7 and delivery of the sealed certification. Construction-phase services under Task 8 extend through completion of construction and final acceptance, estimated at an additional six (6) months, at the University's option.

Durations are the University's planning estimate and are subject to confirmation by the Office of Procurement.

5 Place of Performance

Engineering work shall be performed at the Contractor's place of business. Site visits shall be performed at the Lake Clifton Aerospace Engine Test Site, Baltimore, Maryland, under §3.7 and §12. Meetings shall be held at the Morgan State University campus or by video conference. The Contractor shall include at least two site visits during design, and the visits needed for Task 8 if exercised.

6 Contractor Qualifications and Key Personnel

The Contractor shall demonstrate in its response that it meets each qualification below. Failure to demonstrate a mandatory qualification may render the response non-responsive.

No.	Qualification	Basis
1	The Engineer of Record shall be a Professional Engineer licensed and in good standing in the State of Maryland at the time the design is sealed, practising in the structural discipline. An engineer licensed in another state may be proposed if Maryland licensure (for example by comity) is obtained before sealing; the proposal shall state the engineer's licensure status and plan.	Mandatory
2	The Engineer of Record shall have at least five (5) years of experience in the design of cast-in-place reinforced-concrete foundations.	Mandatory
3	Demonstrated experience with anchorage to concrete under ACI 318-19 Chapter 17, including post-installed anchors and the application of ICC-ES evaluation reports.	Mandatory
4	Insurance of the types and amounts required by §11.3 and Appendix A, Section H of the Policies, including any professional-liability and XCU requirements confirmed by the Procurement Officer.	Mandatory
5	Capability to prepare a special-inspection program under IBC 2021 Chapter 17 and, if Task 8 is exercised, to perform or administer it in-house or through an ASTM C1077-accredited testing agency named in the response.	Mandatory
6	Access to a Maryland-licensed geotechnical engineer, in-house or as a named subconsultant, for Task 3.	Mandatory
7	Experience with spill containment, underground tank installation or stormwater and environmental compliance in Maryland, in-house or through a named subconsultant, for Task 6.	Mandatory
8	Experience with foundations under dynamic, impulsive or vibratory loading, or with propulsion test, launch or similar special facilities.	Desirable
9	Experience with construction on uncontrolled or undocumented fill.	Desirable

The Contractor shall name the Engineer of Record and each subconsultant in its response and shall not substitute them without the University's prior written approval.

7 Tasks

All work shall conform to the codes and standards of §15 and to the standard of care of a licensed professional engineer practising in Maryland. §3.3 to §3.7 apply to every task.

7.1 Task 1 — Project initiation and design review

- Attend a kickoff meeting within ten (10) calendar days of the NTP.

- Review the design, calculations, drawings and assumptions in Appendices A and D and the information listed in §10.
- Independently verify the load derivation, the distribution of tension to the anchor groups, and every anchor, stability and concrete check in Appendix A.
- Issue a written design review memorandum stating each error, omission or unconservative assumption found and its recommended resolution.
- Confirm or revise the concrete mix requirements (exposure class F2), reinforcement, cover and foundation geometry.

7.2 Task 2 — Wind load analysis

- Determine wind loads on the test stand (about 19.3 ft tall) and attached equipment to ASCE/SEI 7-22 for the site, stating risk category, exposure category and basic wind speed.
- Develop the governing load combinations with wind for the stand at rest and evaluate overturning, sliding, bearing and anchor demand.
- Recommend an operational wind limit for test firings.
- Where wind governs, revise the foundation and anchorage design and document the revision.

7.3 Task 3 — Geotechnical review and foundation bearing

- Review the existing geotechnical information for the site, which describes uncontrolled fill of variable depth (about 0.75 ft to 22 ft) beneath asphalt paving.
- Establish the allowable bearing pressure, total and differential settlement, the base friction coefficient (0.40 assumed in Appendix A) and the design groundwater level.
- Specify subgrade preparation, including removal of asphalt and unsuitable material and replacement with compacted engineered fill or aggregate.
- Evaluate the effect of the containment-tank excavation, which lies within the zone of influence of the pad edge in the preliminary layout (Appendix D, Figure D-2), and specify backfill, a minimum offset or a relocation of the tank so that the pad bearing is not undermined.
- State whether the existing information is sufficient for certification. If it is not, describe the additional investigation and price it as an option in Appendix C. Any drilling or excavation by the Contractor or its subconsultant is subject to the XCU requirement of §11.3.1.

7.4 Task 4 — Anchorage verification

- Verify the anchorage for all applicable ACI 318-19 Chapter 17 failure modes: steel tension and shear, concrete breakout in tension, pullout, side-face blowout, concrete edge breakout in shear, pryout, and tension–shear interaction.
- Confirm that the Hilti HDA-TR M12x125/30 stainless anchor (ICC-ES ESR-1546) is qualified for the application and that the design values used match the report and the manufacturer's printed installation instructions.
- ESR-1546 §5.10 places fatigue and shock loading outside the scope of the report. State the basis on which the anchor is accepted for repeated rocket-firing loads (Appendix A.4.7 bounds the dynamic effect with DAF = 2.0), or specify an alternative.
- Confirm embedment, spacing, gauge, edge distance and pad thickness, and that these are consistent across the PWS, calculations and drawings.
- Demonstrate a demand-to-capacity ratio not exceeding 1.00 for every strength check.

7.5 Task 5 — Baseplate connection, flame deflector and thermal protection

- Check the baseplate connection for plate bending, weld capacity and prying with the baseplates bearing directly on the concrete. Specify the baseplate thickness (15 to 30 mm for HDA-TR M12x125/30) and the hole diameter (22 to 23 mm) that the Stand Fabricator must provide. Any modification of the stand is carried out by others.
- Review the flame deflector geometry and design its anchorage to the pad for the jet reaction (of the order of the thrust), with deflector anchors outside the projected breakout areas of the stand anchors, and confirm global sliding.
- Review the thermal protection (deflector with a carbon-steel shield plate on standoffs) for firings within the 30-second envelope, and state inspection and replacement criteria for the shield plate.
- Confirm the minimum separation between the deflector exhaust and the trench drain, grate and any polymer components, or specify a formed-concrete trench where that separation cannot be met.

7.6 Task 6 — Drainage and spill-containment design

The preliminary arrangement is described in Appendix A.6 and shown in Appendix D. The Contractor shall verify or revise it and design it for construction so that it meets the following performance requirements.

- **Level deck.** The 8 × 8 ft deck under the stand shall be level (flatness ACI 117 F_F 35 or better) so that the stand is plumb and every baseplate bears fully on the concrete.
- **Collection.** All quench water, deluge water, wash water, rainfall on the pad and spilled RP-1 shall be collected by the sloped apron (preliminary: 18 in wide at ½ in per ft) and the trench drain and conveyed to containment. A containment curb shall bound the remaining sides.
- **Hydraulic capacity.** The apron, trench, grate and outlet shall carry the larger of the fire-protection water-spray flow under PWS-LC-FPE-001 and the design rainfall intensity for the site, with stated freeboard. The preliminary design uses 50 gpm.
- **Containment volume.** The buried tank, together with any stated deck retention, shall hold not less than the sum of 110 % of the largest volume of RP-1 that can be present on the pad, the water-spray volume over the 30-second envelope plus post-test wash-down, and the design storm runoff from the pad between pump-outs. The preliminary nominal size is 300 gallons.
- **Tank construction.** Double-wall tank listed for flammable and combustible liquids (steel to UL 58 with UL 1746 corrosion protection, or fiberglass to UL 1316), with interstitial leak monitoring, vent, pump-out riser, isolation valve, and bonding and grounding of the grate, piping and tank.
- **Flotation.** Hold-down slab or deadmen giving a factor of safety of at least 1.2 against flotation of the empty tank at the design groundwater level established in Task 3.
- **No discharge.** No connection to the storm drain or to a soak-away. Contents shall be pumped out and disposed of through the University's hazardous-waste program.
- **Material compatibility and temperature.** All wetted components compatible with RP-1, and all components protected from the exhaust and radiant heat (Task 5).
- **Regulatory determination.** Identify whether the tank and the pad are regulated (including MDE COMAR 26.10, NFPA 30, Baltimore City requirements and stormwater rules), list every permit, registration or approval required, and prepare the design documentation the University needs to apply. Coordinate with the Office of Safety, Health and Environment and the fire-protection engineer.

7.7 Task 7 — Certified design package

- Prepare a complete calculation package incorporating Tasks 1 through 6.
- Prepare construction drawings: foundation plan, sections, reinforcement with anchor clearance zones, anchor layout and details, deflector anchorage, thermal protection, drainage, trench, containment tank, hold-down and piping.
- Prepare specifications for concrete (exposure class F2), reinforcement, anchor installation per ESR-1546, drainage and tank installation, testing and tolerances. The specifications shall place on the Construction Contractor the obligations of §3.6 and §3.7.
- Prepare the statement of special inspections to IBC 2021 Chapter 17, identifying each item, its frequency and its acceptance criteria.
- Sign and seal the drawings, calculations and specifications, and execute a written certification that the design is adequate for the loads and conditions stated.

7.8 Task 8 — Construction-phase engineering services (optional)

Services under this task are limited to review, observation, inspection and certification and are subject to §3.4. The Contractor shall not direct, supervise or control the Construction Contractor's operations.

- Review submittals (mix design, reinforcement shop drawings, anchor data and installation procedures, trench and tank data) for conformance with the sealed design.
- Respond to requests for information within five (5) business days.
- Perform or administer special inspections, including reinforcement scanning before drilling, anchor hole preparation, embedment, sleeve recess and installation torque, concrete testing, and tank installation and hold-down.
- Perform a final site observation and issue a written statement that the constructed work, as observed and inspected, conforms to the sealed design.
- Deliver the complete record set.

8 Deliverables and Delivery Schedule

Deliverables shall be submitted as searchable PDF and, on request, in native format. Sealed documents shall also be provided as signed and sealed originals. Days are calendar days from the NTP. Each deliverable is priced separately in Appendix C.

No.	Deliverable	Task	Due
D-1	Kickoff meeting and project schedule	1	10 days
D-2	Design review memorandum with findings and resolutions	1	30 days
D-3	Wind load analysis, load combinations and operational wind limit	2	45 days
D-4	Geotechnical review: allowable bearing, settlement, friction, groundwater, subgrade preparation, tank-excavation effect	3	45 days
D-5	Anchorage verification calculations, including acceptance basis for firing loads	4	60 days
D-6	Baseplate requirements for the Stand Fabricator; deflector anchorage; thermal-protection review	5	60 days
D-7	Drainage and spill-containment design, tank sizing and flotation check, and regulatory determination	6	75 days

No.	Deliverable	Task	Due
D-8	Draft (90 %) calculations, drawings and specifications for University review	7	90 days
D-9	Signed and sealed drawings, calculations and specifications	7	120 days
D-10	Written certification and statement of special inspections	7	120 days
D-11	Submittal reviews and RFI responses (optional)	8	During construction
D-12	Special inspection reports (optional)	8	5 business days after each inspection
D-13	Final conformance statement and record set (optional)	8	30 days after construction completion

9 Performance Requirements Summary

Required service	Performance standard	Acceptable quality level	Surveillance
Sealed certification	Sealed by a Maryland-licensed PE; every failure mode documented with DCR ≤ 1.00 and every stability FS at or above its stated minimum	100 %	University technical review
Code compliance	Conforms to the codes of §15, each governing provision cited	100 %	Calculation review
Environmental performance	Containment design meets every requirement of Task 6; no discharge path to ground or storm drain	100 %	Review of D-7 and drawings
Closure of open items	Every Appendix B item closed and documented	100 %	Item-by-item review
Timeliness	Deliverables on or before the §8 dates	95 % on time	Schedule tracking
Technical accuracy	No errors requiring re-issue of the sealed package	≤ 1 re-submittal per deliverable	Technical review
Responsiveness	RFIs answered within 5 business days	90 %	RFI log
Coordination	No disruption of site operations or adjacent work (§3.7)	100 %	PI and Project Manager reports

10 University-Furnished Information and Services

Furnished at no cost and not warranted by the University; the Contractor shall verify any furnished information it relies upon.

- This PWS, including Appendices A, B and D.
- The existing geotechnical report for the Lake Clifton site.
- **The steel test stand, as owner-furnished equipment**: structural design, CAD models, drawings, baseplate geometry, leg spacing and weights, as design data only.
- Engine operating parameters: thrust, propellants, on-stand quantities and firing durations.
- ICC-ES ESR-1546 and the preliminary anchor calculations.

- PWS-LC-FPE-001 Revision 1 and, when available, the fire-protection design deliverables, including water-spray flows.
- Site access for site visits and, if Task 8 is exercised, for construction-phase services.
- A single point of contact for technical coordination (§16).

11 Contractor-Furnished Items, Insurance and Risk Allocation

11.1 Contractor-furnished items

Except for the items in §10, the Contractor shall furnish all personnel, licensed engineering services, subconsultants, software, transportation, equipment and incidentals needed to perform §7 and produce §8.

11.2 Governing procurement policy and acceptance of terms

All procurements of Morgan State University are conducted on behalf of the University in accordance with the **Morgan State University Procurement Policies and Procedures** (approved by the Maryland Board of Public Works, 21 September 2005, as amended; "the Policies"), available at https://www.morgan.edu/Documents/ADMINISTRATION/OFFICES/procurement/Policies_Procedures.pdf. This solicitation and any contract resulting from it are governed by the Policies. The Uniform Contract Terms and Conditions set out in Appendix A of the Policies are incorporated into this PWS by reference and govern over this PWS in the event of any conflict.

By submitting a quotation, bid or proposal in response to this solicitation, the offeror accepts the University's terms and conditions as set out in the Policies and in this PWS, in full and without addition or modification, and agrees that any resulting award will be made subject to them. Any proposal of terms in addition to or different from those of the University shall be treated in accordance with Appendix A, Section D, item 24 (Conflicting Terms) of the Policies.

The following provisions of Appendix A, Section D of the Policies (pp. 60–65) apply to the resulting contract and are quoted verbatim:

2. Incorporation by Reference: "All terms and conditions of the solicitation, and any changes thereto, are made a part of this contract."

20. Maryland Law Prevails: The provisions of this contract shall be governed by the laws of Maryland.

22. Pre-existing Regulations: The regulations set forth in MORGAN STATE UNIVERSITY Procurement Policies and Procedures in effect on the date of execution of this Contract are applicable to this Contract.

Appendix A, Section E, item 22 of the Policies (Compliance with Laws, p. 82) is a mandatory provision for all contracts and is quoted verbatim:

"The Contractor hereby represents and warrants that:

a. It is qualified to do business in the State of Maryland and that it will take such action as, from time to time hereafter, may be necessary to remain so qualified;

b. It is not in arrears with respect to the payment of any moneys due and owing the State of Maryland, or any department or unit thereof, including but not limited to the payment of taxes and employee benefits, and that it shall not become so in arrears during the term of this Contract;

c. It shall comply with all federal, State, and local laws, regulations, and ordinances applicable to its activities and obligations under this Contract; and

d. It shall obtain, at its expense, all licenses, permits, insurance, and governmental approvals, if any, necessary to the performance of its obligations under this Contract."

Procurement method

The Office of Procurement will determine the procurement method and state it in the solicitation: a Request for Proposals with Best Value evaluation under State Finance and Procurement Article § 11-203, a small procurement under COMAR 21.05.07 and the Policies, the qualification-based architectural and engineering procedure of Section V.I of the Policies, or an intergovernmental cooperative purchase. **Whatever the method, each offeror shall complete the itemized price schedule of Appendix C**, so that the University can evaluate price reasonableness deliverable by deliverable. If the contract value exceeds $100,000, the Truth-in-Negotiation Certification of Appendix A, Section E, item 24 of the Policies will also apply.

11.3 Insurance

The Contractor shall maintain the insurance required by Appendix A, Section H of the Policies. Section H states that its terms are for use in contracts where purchase of insurance by the contractor is required by law, regulation or the procurement officer's judgment, and that the terms and amounts of the clause may be changed at the procurement officer's discretion. Items 1 through 5 of Section H are quoted verbatim below and form part of this contract; the University's applicability determinations for this contract follow in §11.3.1.

Morgan State University Procurement Policies and Procedures, Appendix A, Section H — Insurance (pp. 109–110), quoted verbatim:

1. The Contractor shall defend, indemnify and save harmless Morgan State University, its officers, employees and agents, from any and all claims, liability, losses and causes of actions which may arise out of the performance by the Contractor, employees or agents, of the work covered by this Contract.

2. The Contractor shall secure, pay the premiums for, and keep in force until the expirations of this Contract, and any renewal thereof, adequate insurance as provided below, such insurance to specifically include liability assumed by the Contractor under this Contract.

(a) Commercial General Liability Insurance including all extensions — $2,000,000 each occurrence; $2,000,000 personal injury; $2,000,000 products/completed operations; $2,000,000 general aggregated.

(b) Workmen's Compensation Insurance and Unemployment Insurance as required by the laws of the State of Maryland.

(c) Owner's Landlord's and Tenant's and Contractor's bodily injury liability insurance, with limits of not less than $500,000 for each person and $2,000,000 for each accident.

(d) Property damage liability insurance with a limit of not less than $2,000,000 for each accident.

(e) If automotive equipment is used in the operation, automobile bodily injury liability insurance with limits of not less than $1,000,000 for each person and $2,000,000 for each accident, and property damage liability insurance, with a limit of not less than $2,000,000 for each accident.

(f) Food products liability insurance, if not included in the Comprehensive, with limits of not less than $1,000,000 for each person and $2,000,000 for each accident.

3. All policies for liability protection, bodily injury or property damage must specifically name or its face, Morgan State University as an additionally named insured as respects to operations under the contract and premises occupied by the Contractor provided, however, with respect to the Contractor's liability for bodily injury or property damage under items 2a-2f above, such insurance shall cover and not exclude Contractor's liability for injury to the property of the University System and to the persons or property of employees, students, faculty members, agents, officers, regents, invitees or guests of the University System.

4. Each insurance policy shall contain the following endorsements: "It is understood and agreed that the Insurance Company shall notify in writing Director of Procurement or an authorized designee forty-five (45) days in advance of the effective date of any reduction in or cancellation of this policy." A certificate of each policy of insurance shall be furnished to the Director of Procurement or an authorized designee. With the exception of Workmen's Compensation, upon the request of the Director of Procurement or an authorized designee a certified true copy of each policy of insurance, including the above endorsement manually countersigned by an authorized representative of the insurance company, shall be furnished to the Director of Procurement or an authorized designee. A certificate of insurance for Workmen's Compensation together with a properly executed endorsement for cancellation notice shall also be furnished. Following the notice of Contract award, the requested Certificates and Policies shall be delivered as directed by the Director of Procurement or an authorized designee. Notices of policy changes shall be furnished to the Director of Procurement or an authorized designee.

5. All required insurance coverage must be acquired from insurers allowed to do business in the State of Maryland and acceptable to the University. The insurers must have a policyholders' rating of "A-" or better, and a financial size of "Class VII" or better in the latest edition of Best's Insurance Reports.

11.3.1 Applicability to this contract

- Items 2(a) to 2(d) and items 3, 4 and 5 apply in full. Item 2(e) applies if automotive equipment is used, including travel to the site. Item 2(f) does not apply.
- Certificates meeting item 4 shall be delivered before the NTP, and coverage maintained for the full period of performance, including any exercised option.
- **Explosion, collapse and underground (XCU).** If the Contractor or any subconsultant performs drilling, test pits or other subsurface work at the site, its commercial general liability insurance shall not exclude explosion, collapse or underground property damage hazards, and Morgan State University shall be named additional insured for that work. The University will require the same XCU coverage in the construction contract for excavation of the pad and the containment tank.
- **Professional liability (proposed — for decision by the Procurement Officer under Section H).** Because the deliverable is a sealed engineering design, the requesting department recommends professional liability (errors and omissions) insurance of not less than $1,000,000 per claim and $2,000,000 aggregate, maintained for at least three (3) years after final acceptance.

11.4 Indemnification

The Contractor's obligation to defend, indemnify and save harmless the University is item 1 of Appendix A, Section H of the Policies, quoted in §11.3 above. The following provisions of Appendix A, Section D of the Policies (pp. 64–65) also apply and are quoted verbatim:

19. Intellectual Property: Contractor agrees to indemnify and save harmless the University, its officers, agents and employees with respect to any claim, action, cost or judgment for patent infringement, or trademark or copyright violation arising out of purchase or use of materials, supplies, equipment or services covered by this contract.

23. Indemnification: The University shall not assume any obligation to indemnify, hold harmless, or pay attorneys' fees that may arise from or in any way be associated with the performance or operation of this agreement.

Nothing in §3.4 or elsewhere in this PWS limits the Contractor's obligations under these provisions.

11.5 Bonds

Appendix A, Section G — Bonds (p. 99) of the Policies provides, verbatim:

A bid bond, performance bond, or a payment bond may be required in a solicitation. A bid bond, when specified, must accompany the bid. Performance bonds and payment bonds, if requested, must be filed in accordance with the direction of the Director of Procurement or an authorized designee.

No bid bond, performance bond or payment bond is required by this solicitation for the professional engineering services procured by this PWS. Under Appendix A, Section F, item 11 of the Policies, quoted below, performance and payment bonds are mandatory only for construction contracts exceeding $100,000. That requirement will apply to the separately procured construction contract for the foundation and is not a term of this contract.

Appendix A, Section F, item 11 — Performance and Payment Bonds (p. 99), quoted verbatim:

Mandatory provision for all construction contracts exceeding $100,000:

A. Performance Bond. The required performance Bond shall be in the form specified in Section G.1 of these Uniform Contract Terms and Conditions.

B. Payment Bond. The required payment bond shall be in the form specified in Section G.2 of these Uniform Contract Terms and Conditions.

11.6 Solicitation notices required by the Policies

The following notices are mandatory provisions of all University solicitations under Appendix A, Section B of the Policies (pp. 49–50) and are included here verbatim so that they carry into the solicitation without alteration.

Public Information Act Notice (B.5): Offerors should give specific attention to the identification of those portions of their responses that they deem to be confidential, proprietary information or trade secrets and provide any justification why such materials, upon request, should not be disclosed by the University under the Access to Public Records Act, State Government Article, Title 10, Subtitle 6, Annotated Code of Maryland.

Minority Business Enterprise Notice (B.7): "Minority business enterprises are encouraged to respond to this solicitation".

Arrearages (B.8): "By submitting a response to this solicitation, a vendor shall be deemed to represent that it is not in arrears in the payment of any obligation due and owing the State of Maryland, including the payment of taxes and employee benefits, and that it shall not become so in arrears during the term of the contract if selected for contract award."

Bid or Proposal Affidavit (B.9): Each offeror shall complete and submit with its response the Bid or Proposal Affidavit prescribed in Appendix A of the Policies.

Section B.6 of Appendix A (Bid/Proposal Security) applies only where the Procurement Officer elects to require bid security for a contract expected to exceed $100,000; no bid security is required by this PWS (§11.5). The name and telephone number of the Procurement Officer, required by Appendix A, Sections A.4 and B.1, are given in §16.

12 Site Access, Safety and Security

- The site is a former school property with existing asphalt paving and other University activity. The Contractor shall arrange every site visit with the University point of contact at least two (2) business days in advance (§3.7).
- On University property the Contractor shall comply with OSHA requirements and with Morgan State University safety, security and environmental policies, including those administered by the Office of Safety, Health and Environment.
- The Contractor shall provide personal protective equipment for its own personnel and subconsultants.
- No engine test firing shall occur during any Contractor site visit. The Contractor shall not handle propellants or enter propellant storage areas.
- The Contractor shall report any observed unsafe condition or environmental release to the University point of contact immediately.
- The Contractor shall leave the site, existing facilities and adjacent work undisturbed and shall repair at its own cost any damage it causes.
- **These obligations apply to the Contractor's own personnel and subconsultants.** Construction site safety is the Construction Contractor's responsibility (§3.4, §3.6).

13 Meetings, Reporting and Communication

- Kickoff meeting within ten (10) calendar days of the NTP.
- Design review meetings at 30 %, 60 % and 90 % of the design package, at minimum, with one joint coordination meeting with the fire-protection engineer.
- Monthly written progress reports: work completed, work planned, deliverable status, and any issue affecting cost, schedule or adjacent work.
- A request-for-information log submitted with each monthly report.
- All formal correspondence and deliverables to the Procurement Officer (§16), with a copy to the technical point of contact.

14 Acceptance and Payment

A deliverable is accepted when the University has reviewed it and issued written acceptance through the Office of Procurement, on the technical recommendation of the technical point of contact. The

University will respond to each deliverable within ten (10) business days. D-10 will be accepted only when the certification is sealed by a Maryland-licensed Professional Engineer and every Appendix B item is closed.

The Contractor will be paid the price stated in Appendix C for each deliverable after its written acceptance. Options are paid only if exercised in writing by the Office of Procurement. Invoices shall identify the deliverable number and the accepted date.

15 Applicable Codes and Standards

The Contractor shall apply the current adopted editions and state in the calculation package the edition used.

- ACI 318-19 (Chapters 13, 17, 19, 20, 22, 26); ACI 301; ACI 117.
- ASCE/SEI 7-22.
- International Building Code 2021, including Chapter 17, as adopted by Maryland and Baltimore City.
- AISC 360, for the baseplate connection and deflector anchorage.
- ICC-ES ESR-1546 (Hilti HDA, reissued March 2026, revised April 2026) and the manufacturer's printed installation instructions.
- NFPA 30, Flammable and Combustible Liquids Code; NFPA 55 and NFPA 15 as they bear on the coordination with PWS-LC-FPE-001.
- COMAR 26.10 (Oil Pollution and Tank Management) and other Maryland Department of the Environment regulations, as determined applicable in Task 6.
- UL 58, UL 1746 and UL 1316, as applicable to the containment tank.
- ASTM A36, A615, and C31, C39, C143, C231, C1064 and C1077, as applicable.
- COMAR and the rules of the Maryland State Board for Professional Engineers governing the practice of engineering.

16 Points of Contact

Role	Name / Office	Telephone	Email
Procurement Officer (contracting authority)	Oluwagbenga Alo, Office of Procurement	443-885-3176	oluwagbenga.alo@morgan.edu
Principal Investigator and technical point of contact	Dr. Oluwatobi Busari, School of Engineering	443-885-4237	oluwatobi.busari@morgan.edu
Project Manager	Angelo Pirali		angelo.pirali@morgan.edu
Requestor	Mazen Badr, Aerospace Engineering	443-442-7544	mabad5@morgan.edu
MBE Liaison Officer	Karen Reyes		

The Procurement Officer is the University's sole contracting authority. The technical point of contact coordinates and reviews the Contractor's work but has no authority to change the contract, its price, scope or period of performance, to accept deliverables except by written recommendation to the Procurement Officer, or to direct the Construction Contractor or the Stand Fabricator.

All questions about this solicitation shall be directed in writing to the Procurement Officer. Offerors shall not contact technical personnel directly during the solicitation period.

17 Attachments

- Appendix A — Technical Design Basis and Calculations.
- Appendix B — Open Items to be Closed by the Contractor.
- Appendix C — Itemized Price Schedule.
- Appendix D — Preliminary Drawings.
- Attachment 1 — Geotechnical report for the Lake Clifton site (furnished on request).
- Attachment 2 — Test-stand drawings and CAD model (furnished on request).
- Attachment 3 — ICC-ES ESR-1546 (furnished on request).
- Attachment 4 — Morgan State University Procurement Policies and Procedures, Appendix A (incorporated by reference, §11.2).
- Attachment 5 — Morgan State University Construction General Conditions (2022) (to be attached by the Office of Procurement).
- Attachment 6 — PWS-LC-FPE-001 Revision 1, Fire Protection Engineering Services (for coordination).

Appendix A — Technical Design Basis and Calculations

This appendix is the University's preliminary design (technical basis Rev. 0A of 29 September 2026, with the drainage and containment design added in Rev. 3). It is furnished for information. It is not a certified design. The Contractor shall verify every value and shall not rely on it without verification; where the Contractor's analysis differs, the Contractor's sealed result governs.

A.1 Geometry, materials and specification

Parameter	Preliminary value	Basis / note
Concrete pad	8 ft × 8 ft × 15 in, top level (datum 0.00)	Dead weight for stability; exceeds HDA minimum member thickness
Concrete	f′c = 5,000 psi at 28 days, normal weight	Within ESR-1546 range 2,500–8,500 psi
Exposure class	F2	ACI 318-19 Table 19.3.1.1 (outdoor, frequent wetting incl. deluge)
Max w/cm; air	0.45; 6.0 % ± 1.5 % (¾ in aggregate)	ACI 318-19 Tables 19.3.2.1 and 19.3.3.1
Reinforcement	#5 at 9 in each way, top and bottom mats, ASTM A615 Gr. 60	Uplift reverses the usual bending sense
Cover	3 in bottom and sides; 2 in top	ACI 318-19 Table 20.5.1.3.1
Anchors	16 × Hilti HDA-TR M12x125/30, stainless (HDA-R), through-set	ICC-ES ESR-1546; stainless required outdoors (§5.14)
Effective embedment	h_ef = 125 mm (4.92 in)	ESR-1546 Table 5
Arrangement	4 groups of 4 on a 4 in square; groups on a 5.25 ft (63 in) square	Matches the stand; 4 in ≥ s_min = 3½ in
Edge distance	c_a1 = 14.5 in minimum	> 1.5 h_ef = 7.4 in: no edge reduction in tension
Baseplate holes	22–23 mm (0.87–0.91 in); existing ≈0.70 in holes to be enlarged	ESR Table 2 maximum clearance hole for HDA-T M12
Baseplate thickness	15–30 mm (/30) or 20–50 mm (/50)	ESR Tables 1 and 6; open item B-1
Baseplate bearing	Directly on concrete; no leveling nuts or grout	ESR Figure 7, step 1
Installation	Stop drill bit TE-C/TE-Y-HDA-B 22x155; setting tool HDA-ST M12; torque 80 N·m (59 ft-lb) max; sleeve recess 2–7 mm	ESR-1546 Tables 2 and 4A–4C
Stand and rocket	1,500 lb and 330 lb	University CAD model
Subgrade	4–6 in compacted crushed stone over engineered fill; excavation 21 in below pad top	Existing fill 0.75–22 ft; Task 3

A.2 Loads and load combinations

Load	Value	Description
Thrust reaction P	3,000 lbf upward	Maximum engine thrust
Off-axis case θ	7.5°: N = 2,974 lbf, V = 391 lbf	Thrust misalignment
Thrust line height h	8.25 ft above top of pad	From CAD

Load	Value	Description
Dead load	Pad 12,000 lb; stand 1,500 lb; rocket 330 lb; ΣW = 13,830 lb	Apron, curb and trench encasement conservatively ignored
Jet reaction on deflector	≈ P downward; horizontal up to ≈ P	Depends on deflector geometry (B-6)
Wind	To be determined (ASCE 7-22)	Stand at rest; operational limit for firing (B-2)

```
Strength:   U = 1.4 D
Strength:   U = 1.2 D + 1.6 T
Strength:   U = 0.9 D + 1.6 T      <-- governs uplift on the anchorage
Strength:   U = 1.2 D + 1.0 W      (stand at rest; open item B-2)
Strength:   U = 0.9 D + 1.0 W      (stand at rest; open item B-2)
Stability:  service D versus service thrust, compared to a required FS
```

T is the thrust effect treated as a live load. 0.9D + 1.6T is an engineering combination for an operational load that counteracts dead load and is to be accepted by the Engineer of Record. The anchor demands below omit the dead load. Wind is not combined with thrust at design wind speed, because firings are not conducted in design-level wind.

A.3 Requirements and constraints matrix

Failure mode	Demand	Limit (φ included)	ACI 318-19	DCR / FS	Status
Anchor steel — tension	430 lb / anchor	φN_sa = 11,397 lb	§17.6.1	3.8 %	PASS
Concrete breakout — tension	1,720 lb / group	φN_cbg = 19,442 lb	§17.6.2	8.8 %	PASS
Pullout	—	Does not control (ESR §4.1.4)	§17.6.3	—	N/A
Anchor steel — shear	40 lb / anchor	φV_sa = 5,845 lb	§17.7.1	0.7 %	PASS
Edge breakout — shear	160 lb / group	φV_cbg = 17,010 lb	§17.7.2	0.9 %	PASS
Pryout	160 lb / group	φV_cpg = 41,874 lb	§17.7.3	0.4 %	PASS
Tension + shear	combined	sum ≤ 1.2	§17.8	0.098	PASS
Overturning	M_ot = 15,715 lb·ft	FS ≥ 1.5	Practice	FS 3.52	PASS
Sliding	H = 391 lb	FS ≥ 1.5	Practice	FS 11.1	PASS
Flotation / net uplift	U = 3,000 lb	FS ≥ 1.2	Practice	FS 4.6	PASS
Soil bearing	q_max = 216 psf	≤ allowable	Geotech	—	PASS *
Pad flexure	M_u = 188 lb·ft/ft	φM_n = 20,124 lb·ft/ft	Ch. 22	0.9 %	PASS
Pad one-way shear	V_u = 240 lb/ft	φV_c = 8,014 lb/ft	§22.5	3.0 %	PASS
Service cracking	M_s = 118 lb·ft/ft	M_cr = 19,887 lb·ft/ft	Ch. 19/24	0.6 %	PASS

** Allowable bearing for the existing fill is not yet established (open item B-3). Overturning governs. With every thrust effect doubled (DAF = 2.0), all checks pass and overturning is FS 1.76.*

A.4 Calculations

A.4.1 Load analysis

```
N = P cos θ = 3,000 × cos 7.5° = 2,974 lb (uplift)     V = P sin θ = 391 lb (lateral)
```

```
ΣW = 12,000 (pad 8×8×1.25 ft × 150 pcf) + 1,500 (stand) + 330 (rocket) = 13,830 lb
Uniform uplift share (full P, conservative): 3,000 / 16 = 187.5 lb per anchor
M = V · h = 391 × 8.25 = 3,226 lb-ft = 38,709 lb-in
Σc² = 8(33.5)² + 8(29.5)² = 8,978 + 6,962 = 15,940 in²
Extra tension, extreme anchor: 38,709 × 33.5 / 15,940 = 81.4 lb
Service tension: 187.5 + 81.4 = 268.9 lb
N_ua = 1.6 × 268.9 = 430 lb per anchor;  group of four: 1,720 lb
V_ua = 1.6 × 391 / 16 = 39.1, say 40 lb per anchor;  group: 160 lb
```

A.4.2 Anchor tension (ACI 318-19 §17.6, ESR-1546)

```
Steel:     N_sa = 15,196 lb (ESR Table 5, M12: A_se = 0.131 in², f_uta = 116,000 psi)
           φN_sa = 0.75 × 15,196 = 11,397 lb      DCR = 430 / 11,397 = 3.8 %
Breakout:  N_b = k_cr λ √f'c h_ef^1.5 = 24 × 1.0 × 70.7 × 4.92^1.5 = 18,520 lb
           A_Nc = (3 × 4.92 + 4)² = 351.9 in²;  A_Nco = 9 h_ef² = 217.9 in²;  ratio 1.615
           ψ_ed,N = ψ_ec,N = ψ_c,N = ψ_cp,N = 1.0 (c_a,min 14.5 in > 1.5 h_ef; ESR §4.1.11)
           N_cbg = 1.615 × 18,520 = 29,910 lb;  φN_cbg = 0.65 × 29,910 = 19,442 lb
           DCR = 1,720 / 19,442 = 8.8 %   (governs tension)
Pullout:   does not control for the HDA (ESR §4.1.4).  Side-face blowout: c = 14.5 in ≫ 0.4 h_ef.
```

A.4.3 Anchor shear (ACI 318-19 §17.7)

```
Steel:     V_sa = 8,992 lb (HDA-PR M12, ESR Table 5, lower bound; HDA-TR /30 Table 6 = 20,235 lb)
           φV_sa = 0.65 × 8,992 = 5,845 lb      DCR = 40 / 5,845 = 0.7 %
Edge:      d_a = 0.83 in; l_e = h_ef = 4.92 in; c_a1 = 14.5 in; h_a = 15 in
           V_b = 9 λ √f'c c_a1^1.5 = 35,139 lb (governs over 35,543 lb)
           A_Vc / A_Vco = 603.8 / 946.1;  ψ_ed,V = 0.90;  ψ_h,V = 1.204;  ψ_c,V = 1.0
           V_cbg = 24,300 lb;  φV_cbg = 0.70 × 24,300 = 17,010 lb   DCR = 160 / 17,010 = 0.9 %
Pryout:    φV_cpg = 0.70 × 2.0 × 29,910 = 41,874 lb          DCR = 0.4 %
Interaction (§17.8): 0.088 + 0.009 = 0.098 ≤ 1.2
```

A.4.4 Global stability (service level)

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Overturning:  M_r = ΣW × B/2 = 13,830 × 4.0 = 55,320 lb-ft
              M_ot = P × B/2 + V × (h + t) = 12,000 + 391 × 9.5 = 15,715 lb-ft
              FS = 55,320 / 15,715 = 3.52 ≥ 1.5        <-- governs
Sliding:      μ = 0.40 (to be confirmed);  F_r = 0.40 × (13,830 − 2,974) = 4,342 lb
              FS = 4,342 / 391 = 11.1 ≥ 1.5
Flotation:    FS = 13,830 / 3,000 = 4.61 ≥ 1.2
Bearing:      at rest q = 13,830 / 64 = 216 psf (governs)
              firing: e = 3,715 / 10,856 = 0.34 ft < B/6 = 1.33 ft;  q_max = 213 psf
```

A.4.5 Pad section (full-width strip, per foot)

```
d = 15 − 3 − 0.625 − 0.3125 = 11.06 in;  A_s = 0.413 in²/ft (#5 @ 9 in);  a = 0.49 in
Flexure:  φM_n = 0.9 × 0.413 × 60,000 × (11.06 − 0.24) = 20,124 lb-ft/ft;  M_u = 188  -> 0.9 %
Shear:    φV_c = 0.75 × 8 λ_s λ ρ_w^(1/3) √f'c b d = 8,014 lb/ft (Table 22.5.5.1(c));  V_u = 240  ->
3.0 %
Cracking: M_cr = 7.5√f'c × bh²/6 = 19,887 lb-ft/ft;  M_s = 118  -> 0.6 %  (uncracked in service)
A_s,min = 0.0018 × 12 × 15 = 0.324 in²/ft < 0.413 provided;  ε_t = 0.052 (tension-controlled)
Local concentration bound (×4): flexure 3.7 %, shear 12 %, cracking 2.4 %
```

A.4.6 Flame-deflector jet reaction (bounding)

```
One-sided deflector: H = P + V = 3,391 lb;  normal force = ΣW = 13,830 lb (stand uplift balanced)
F_r = 0.40 × 13,830 = 5,532 lb;  FS = 5,532 / 3,391 = 1.63 ≥ 1.5   (bounding)
A symmetric deflector largely cancels H.  Deflector anchors outside the stand-anchor breakout areas.
```

A.4.7 Dynamic sensitivity (DAF = 2.0)

Check	DAF = 1.0	DAF = 2.0	Limit	Status
Anchor tension (breakout)	8.8 %	17.7 %	≤ 100 %	PASS

Check	DAF = 1.0	DAF = 2.0	Limit	Status
Overturning	FS 3.52	FS 1.76	≥ 1.5	PASS
Flotation	FS 4.61	FS 2.31	≥ 1.2	PASS
Sliding	FS 11.1	FS 4.03	≥ 1.5	PASS
Pad flexure / shear	0.9 % / 3.0 %	1.9 % / 6.0 %	≤ 100 %	PASS

A.5 Thermal protection

- **Flame deflector** (structural): angled steel deflector below the engine turning the plume horizontally away from the pad and from the trench side; anchored for the jet reaction (A.4.6; B-6).
- **Shield plate** (sacrificial, non-structural): ASTM A36, ⅜ to ½ in thick, on standoffs with an air gap, on slotted supports free to expand; inspected after each firing and replaced when warped or scaled.
- **Water cooling**: perforated-pipe spray onto the deflector during sustained firings, as designed under PWS-LC-FPE-001. Its flow sets the drainage design flow (A.6).
- Concrete loses strength and may spall above about 300 °C; exhaust flame temperature is of the order of 3,200 °C. No surface material survives sustained direct impingement, so the plume is turned first and the shield handles residual and radiant heat over the 30-second envelope.

A.6 Drainage and spill containment (preliminary)

Element	Preliminary value	Note
Deck	96 × 96 in, level, F_F 35 finish	Stand plumb; uniform baseplate bearing
Apron	18 in wide, full 8 ft edge, ½ in/ft (4.17 %), 0.75 in fall	Monolithic with pad, both mats carried through; side away from operators
Curb	3 × 4 in, tops at +3.00, three pad sides and apron returns (27 ft)	Deck holds about 158 gal with the outlet valve shut
Trench drain	5.43 × 5.39 in clear, 8 ft-6 in long, 0.5 % fall; galvanized or stainless steel grate ¼ in below apron edge	Encasement 4 in all round, poured with the pad
Outlet	4 in vertical drop from the trench to the tank, with isolation valve	No connection to storm drain or soak-away
Containment tank	Buried double-wall tank, ≈300 gal nominal (≈Ø38 in × 64 in), top about 30 in below grade, under the apron and trench, outside the pad footprint	Pump-out riser and vent to grade; leak monitoring; bonded and grounded
Hold-down	Concrete slab or deadmen under the tank	FS ≥ 1.2 against flotation
Overall footprint	8 ft-0 in × 10 ft-7½ in	Pad + apron + trench
Concrete	≈105.9 ft³ (3.92 yd³); order about 4.25 yd³	Pad 80.0 + apron 14.6 + trench zone 9.0 + curb 2.3 ft³

A.6.1 Hydraulic checks

```
Rainfall:  Q = C i A = 0.95 × 8 in/hr × 76 ft² = 0.0133 cfs = 5.9 gpm
Design flow (deluge / wash-down, preliminary): 50 gpm  <-- governs; confirm with PWS-LC-FPE-001
Apron sheet flow at 50 gpm: depth 0.15 in, velocity 1.2 ft/s
Channel (Manning n = 0.012, 4 in depth, 0.5 % fall): 156 gpm  -> DCR 0.32
Outlet (4 in orifice, Cd 0.60, 4 in head): 109 gpm             -> DCR 0.46 (limits)
```

A.6.2 Containment volume (to be confirmed under Task 6)

```
RP-1 on stand per firing: 38.5 lbm / 6.75 lb/gal            =   5.7 gal
  110 % of on-stand RP-1                                       =   6.3 gal
Water spray, 50 gpm × 30 s envelope                            =  25 gal
Post-test wash-down allowance (10 min at 5 gpm)                =  50 gal
Rainfall on pad + apron, 2 in event: 76 ft² × 2/12 ft          =  95 gal
Sum (single event, tank empty at start)                        ≈ 176 gal
Preliminary tank: 300 gal nominal (about 70 % margin);  deck retention 158 gal additional
Note: RP-1 ready storage (370 lbm ≈ 55 gal) is assumed off the pad; if any container is
      placed on the pad, the Engineer shall add 110 % of its volume.
```

A.6.3 Flotation of the empty tank (indicative)

```
Displaced volume ≈ 300 gal / 7.48 = 40.1 ft³;  buoyancy if submerged = 40.1 × 62.4 ≈ 2,500 lb
Required resisting weight at FS 1.2 ≈ 3,000 lb;  submerged concrete 150 − 62.4 = 87.6 pcf
  -> hold-down concrete of the order of 34 ft³ (≈1.3 yd³) if soil over the tank is ignored
The Engineer shall set the design groundwater level (Task 3) and size the hold-down and straps.
```

> *Preliminary layout issue for Task 3: the tank pit in Figure D-2 lies inside a 1H:1V line drawn down from the bottom edge of the pad base course. The Engineer shall confirm compacted structural backfill or flowable fill, set a minimum offset, or relocate the tank so that the pad bearing is not undermined.*

A.7 Quality assurance and special inspection basis

- Concrete: ASTM C31/C39 cylinders, one set per placement (pad, apron and encasement in one pour); acceptance per ACI 318-19 §26.12.3.1; slump, air (6.0 % ± 1.5 %) and temperature each truck; ASTM C1077 laboratory, ACI-certified technicians.
- Reinforcement: verify bars, both mats, 3 in bottom/side and 2 in top cover; keep top bars at least 1 in clear of the 16 anchor locations without reducing the bar count; photograph before the pour.
- Finish: level and flat at each baseplate (bearing directly on concrete); apron broomed across the fall; water test before final set.
- Anchors: install only after the 28-day strength test is accepted; locate reinforcement by GPR or cover meter before drilling; drill through the positioned baseplate with the stop drill bit; confirm the red setting mark and 2–7 mm sleeve recess; torque to 80 N·m maximum; log recess and torque for every anchor; re-check torque after the first firing.
- Special inspection: periodic, during anchor installation (ESR-1546 §4.4; IBC 2021 §1705.1.1, Table 1705.3), plus reinforcement and concrete placement (IBC §1705.3; ACI 318-19 §26.13), plus tank installation and hold-down as specified by the Engineer.

Appendix B — Open Items to be Closed by the Contractor

No.	Open item	Task	Effect if adverse
B-1	Baseplates for HDA-TR M12: confirm thickness (15–30 mm for /30; 20–50 mm for /50) and 22–23 mm holes; issue the requirement to the Stand Fabricator. If new plates with holes ≤ 14 mm are made, use HDA-PR M12.	5	Moderate — plate rework
B-2	ASCE 7-22 wind on the 19.3 ft stand at rest; operational wind limit for firing.	2	Moderate — may enlarge pad
B-3	Allowable bearing, settlement, subgrade preparation over fill, friction μ = 0.40, design groundwater, and effect of the tank excavation on the pad.	3	Moderate
B-4	Baseplate bending, welds and prying with plates bearing directly on concrete.	5	Moderate — thicker plates
B-5	Fabrication drawing shows a 6 in pad; correct to 15 in on all drawings.	1, 7	High — must be corrected
B-6	Flame deflector geometry and anchorage for the jet reaction; global sliding (one-sided bound FS 1.63); exhaust direction away from the trench.	5	Moderate
B-7	Acceptance of HDA for fatigue and shock (outside ESR-1546 scope, §5.10); DAF = 2.0 bound gives 17.7 %.	4	Low to moderate
B-8	Concrete mix design for exposure F2 (5,000 psi, w/cm ≤ 0.45, air 6.0 % ± 1.5 %).	7	Low
B-9	Drainage design flow from the PWS-LC-FPE-001 water-spray design; hydraulic capacity of apron, trench and outlet.	6	Moderate — larger trench
B-10	Containment tank volume, construction, leak monitoring and hold-down against flotation (FS ≥ 1.2).	6	Moderate
B-11	Regulatory determination for the tank and pad (MDE, NFPA 30, Baltimore City, stormwater) and list of approvals.	6	Moderate — schedule
B-12	Trench length 8 ft-6 in versus 8 ft pad width: confirm the encasement detail at the trench ends.	6	Low
B-13	Seal of a Maryland-licensed Professional Engineer.	7	Required

Appendix C — Itemized Price Schedule

Offerors shall state a firm-fixed price for each line. Payment is made per deliverable on written acceptance (§14). Options are priced separately and paid only if exercised.

Line	Deliverable / service	Basis	Price (USD)
D-1	Kickoff meeting and project schedule	Lump sum	$
D-2	Design review memorandum	Lump sum	$
D-3	Wind load analysis	Lump sum	$
D-4	Geotechnical review (existing information)	Lump sum	$
D-5	Anchorage verification	Lump sum	$
D-6	Baseplate requirements, deflector anchorage, thermal review	Lump sum	$
D-7	Drainage and spill-containment design and regulatory determination	Lump sum	$
D-8	Draft (90 %) package	Lump sum	$
D-9	Sealed drawings, calculations and specifications	Lump sum	$
D-10	Certification and statement of special inspections	Lump sum	$
	Total base price (D-1 to D-10)		$
Opt. 1	Additional subsurface investigation, if required by Task 3 (describe scope)	Lump sum	$
Opt. 2	D-11 Submittal reviews and RFI responses	Lump sum	$
Opt. 3	D-12 Special inspections and testing (state number of visits included)	Per visit / lump sum	$
Opt. 4	D-13 Final conformance statement and record set	Lump sum	$
	Hourly rates for additional services, by labor category (attach)	Rate table	—

Appendix D — Preliminary Drawings

For information only; scaled from the University's preliminary design. Dimensions in inches unless noted.

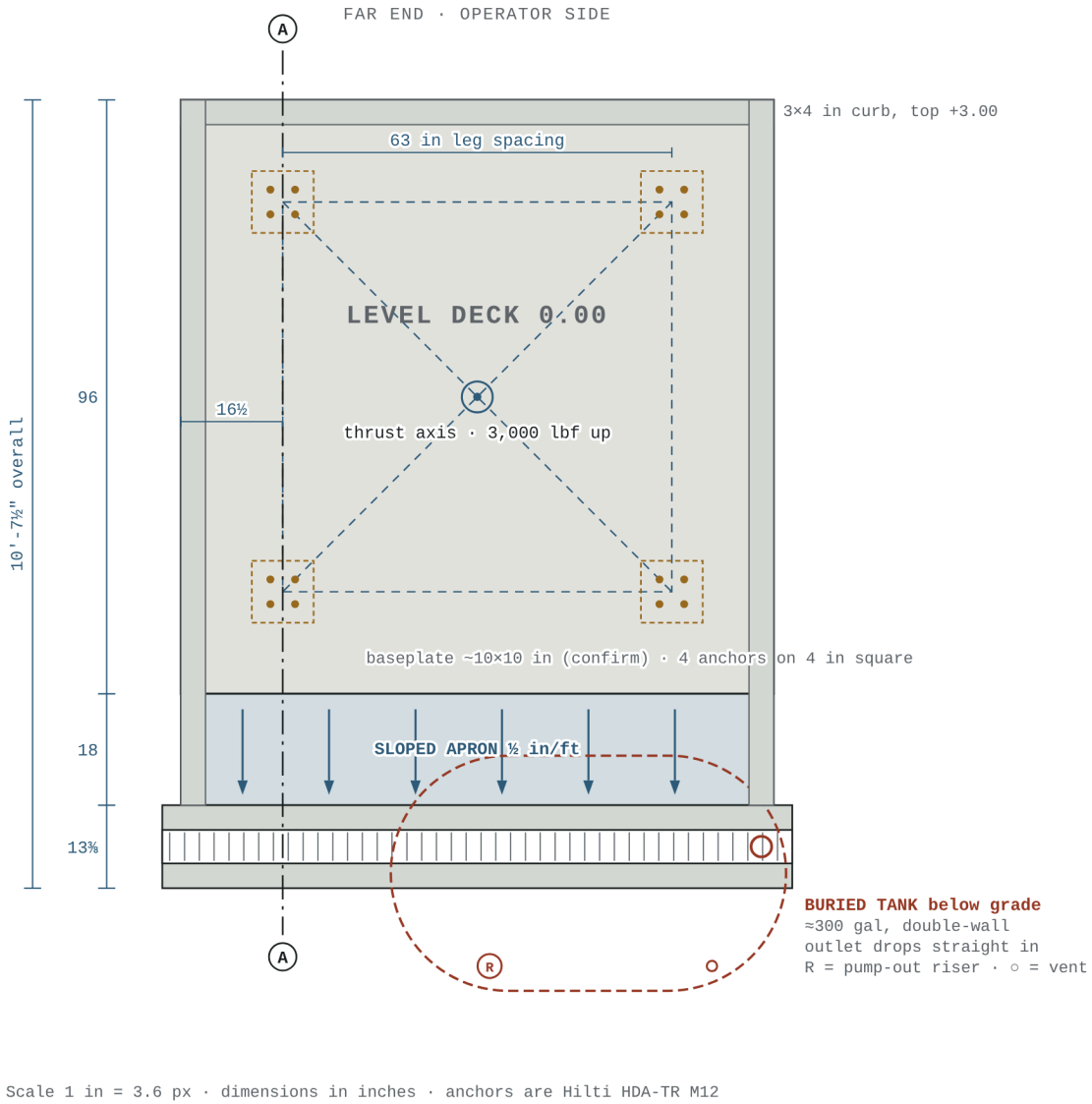


Figure D-1 — Plan: level pad, legs on a 63 in square, 16 anchors, sloped apron, trench drain and buried containment tank (dashed).

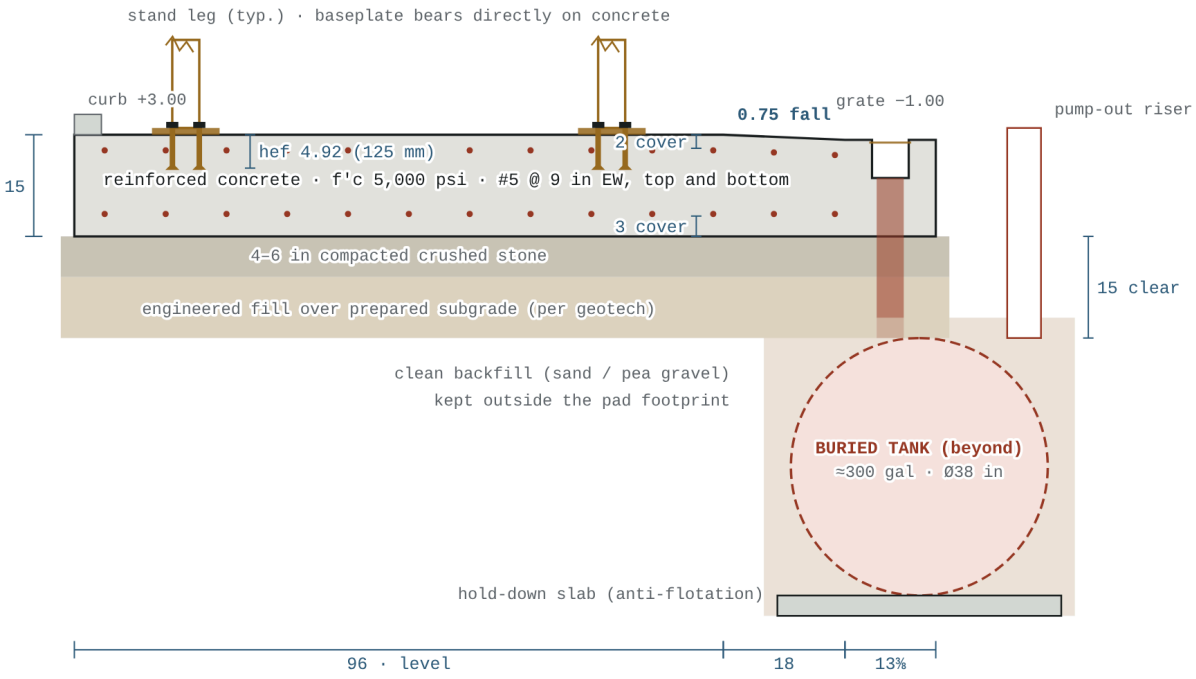


Figure D-2 — Section A–A through the legs: 15 in pad, two mats, HDA anchors at h_ef 125 mm, apron fall 0.75 in, trench, vertical outlet and buried tank (beyond) on a hold-down slab.

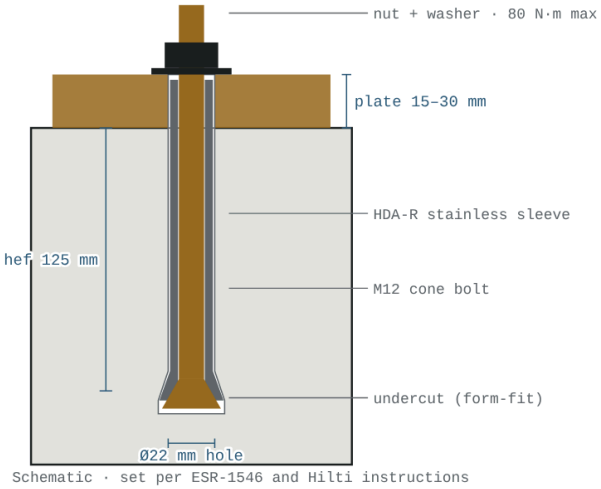


Figure D-3 — Hilti HDA-TR M12 through-set undercut anchor (schematic).